Calculus II Lesson 08

$$\frac{1}{\infty} = \lim_{x\to\infty}\frac{1}{x} = 0 \qquad \frac{-1}{\infty} = \lim_{x\to\infty}\frac{-1}{x} = 0 \qquad \frac{1}{-\infty} = \lim_{x\to-\infty}\frac{1}{x} = 0 \qquad \frac{-1}{-\infty} = \lim_{x\to-\infty}\frac{-1}{x} = 0$$

$$b^{\infty} = \lim_{x\to\infty} b^x = \infty \qquad b^{-\infty} = \frac{1}{b^{\infty}} = \frac{1}{\lim\limits_{x\to\infty} b^x} = \frac{1}{\infty} = 0$$

$$e^{\infty} = \lim_{x\to\infty} e^x = \infty \qquad e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\lim\limits_{x\to\infty} e^x} = \frac{1}{\infty} = 0$$

$$\ln(\infty) = \ln\left(\lim_{x\to\infty} x\right) = \infty$$

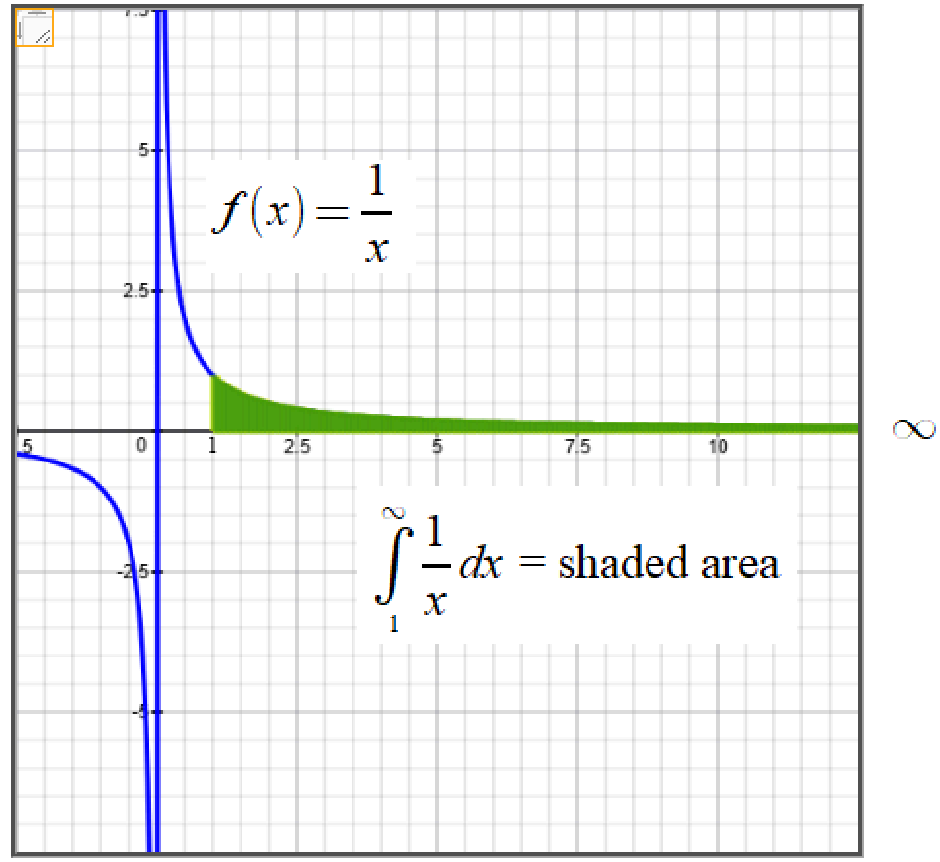
Question 1:

Find $\int_{1}^{\infty} \frac{1}{x}\,dx = \lim_{b\to\infty} \int_{1}^{b} \frac{1}{x}\,dx$

a) The indefinite integral $\int \frac{1}{x}\,dx =$ ___?___

b) $\int_{1}^{\infty} \frac{1}{x}\,dx = \lim_{b\to\infty} \int_{1}^{b} \frac{1}{x}\,dx =$ ___?___

c) Is the shaded area bounded or unbounded?



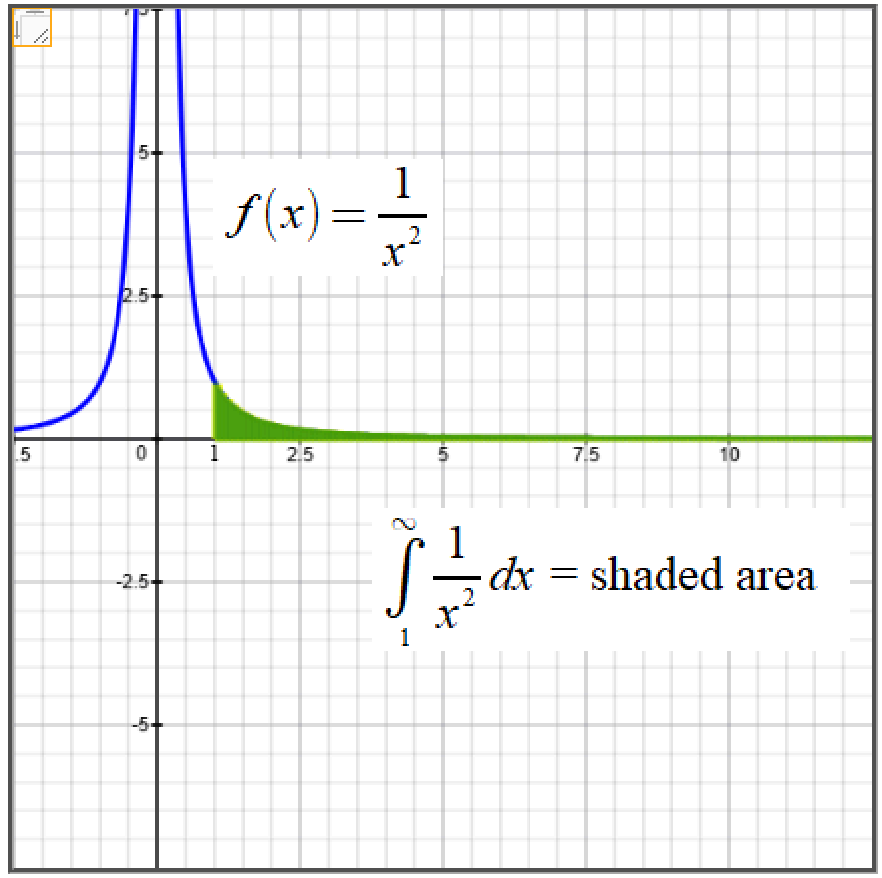
Question 2:

Find $\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^2}\,dx$

a) The indefinite integral $\int \frac{1}{x^2}\,dx =$ ___?___

b) $\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^2}\,dx =$ ___?___

c) Is the shaded area bounded or unbounded?



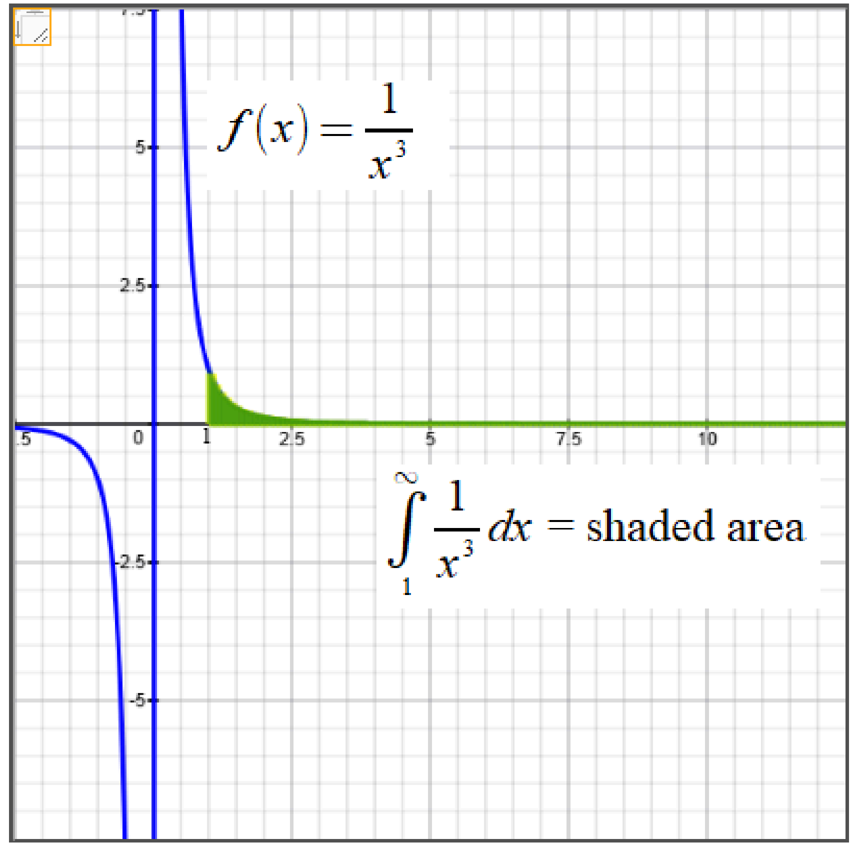
Question 3:

Find $\int_1^\infty \frac{1}{x^3}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^3}\,dx$

a) The indefinite integral $\int \frac{1}{x^3}\,dx = \underline{\quad ? \quad}$

b) $\int_1^\infty \frac{1}{x^3}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^3}\,dx = \underline{\quad ? \quad}$

c) Is the shaded area bounded or unbounded?



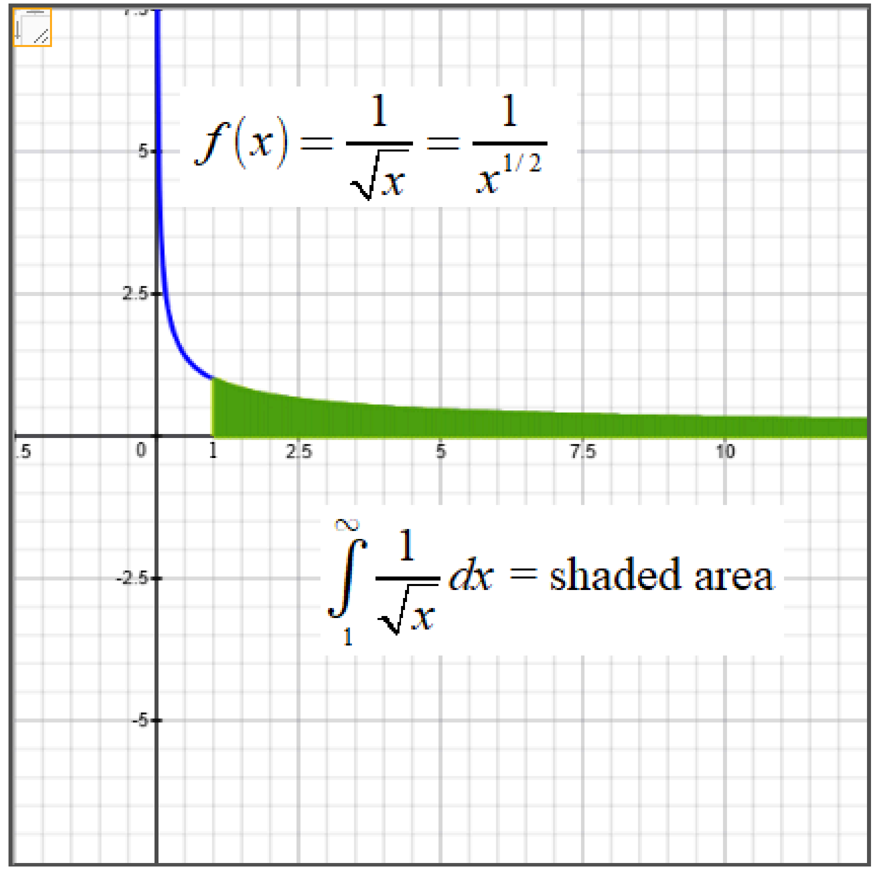
Question 4:

Find $\int_1^\infty \frac{1}{\sqrt{x}}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{\sqrt{x}}\,dx$

a) The indefinite integral $\int \frac{1}{\sqrt{x}}\,dx =$ ___?___

b) $\int_1^\infty \frac{1}{\sqrt{x}}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{\sqrt{x}}\,dx =$ ___?___

c) Is the shaded area bounded or unbounded?



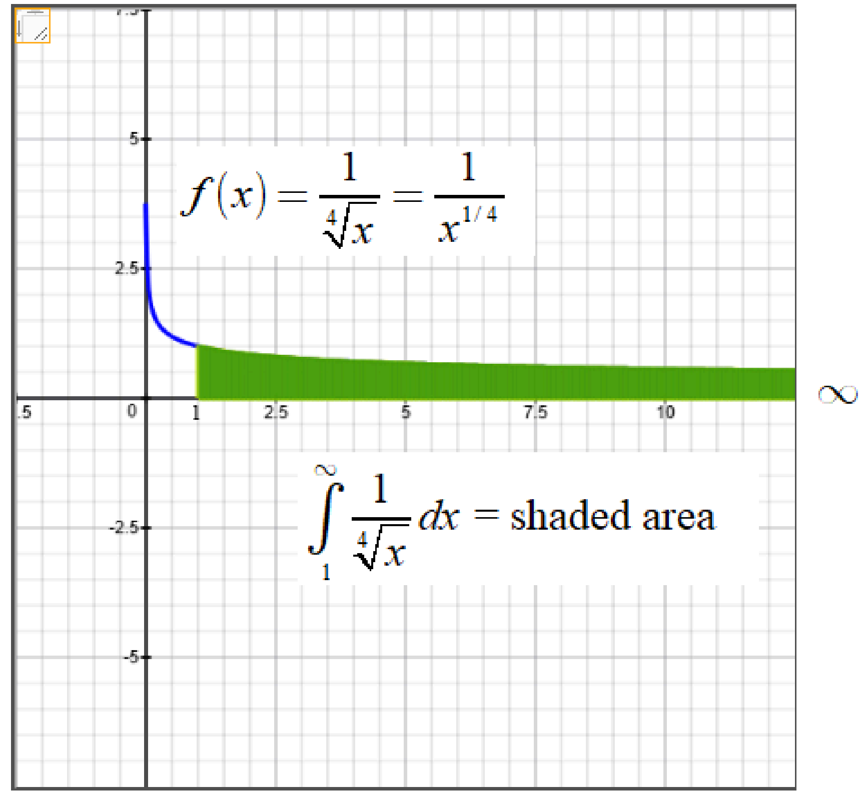
Question 5:

Find $\displaystyle\int_1^{\infty} \frac{1}{\sqrt[4]{x}}\,dx = \lim_{b\to\infty} \int_1^{b} \frac{1}{\sqrt[4]{x}}\,dx$

a) The indefinite integral $\displaystyle\int \frac{1}{\sqrt[4]{x}}\,dx =$ ___?___

b) $\displaystyle\int_1^{\infty} \frac{1}{\sqrt[4]{x}}\,dx = \lim_{b\to\infty} \int_1^{b} \frac{1}{\sqrt[4]{x}}\,dx =$ ___?___

c) Is the shaded area bounded or unbounded?



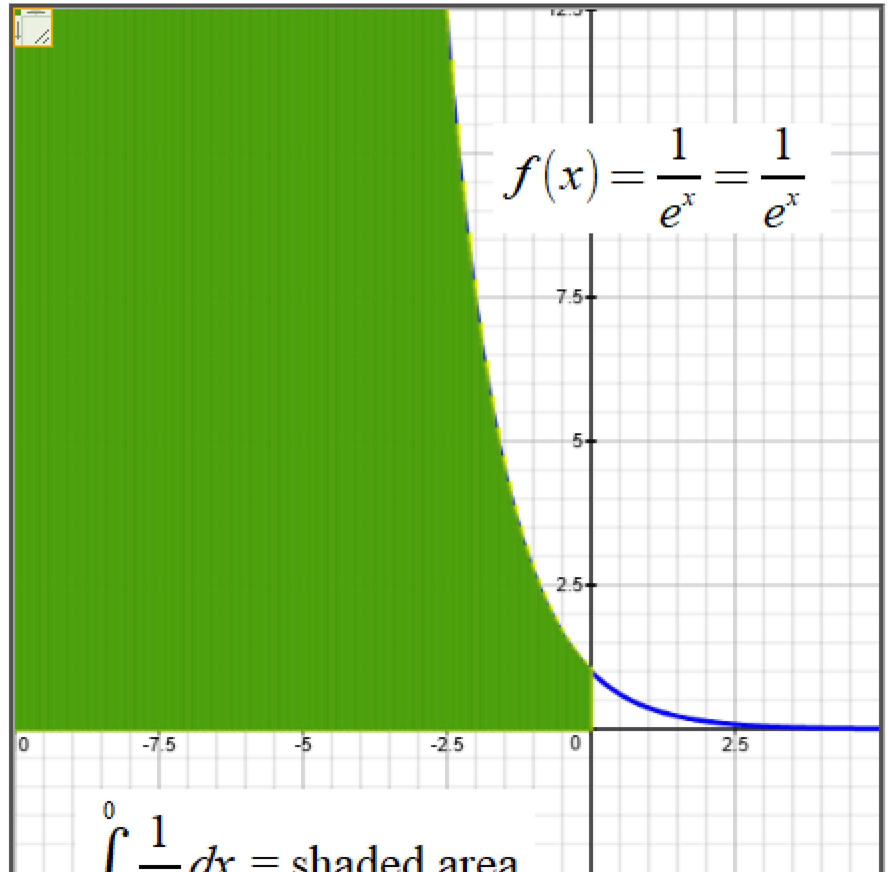
Question 6:

Find $\displaystyle\int_1^\infty \frac{1}{e^x}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{e^x}\,dx$

a) The indefinite integral $\displaystyle\int \frac{1}{e^x}\,dx =$ ___?___

b) $\displaystyle\int_1^\infty \frac{1}{e^x}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{e^x}\,dx =$ ___?___

c) Is the shaded area bounded or unbounded?

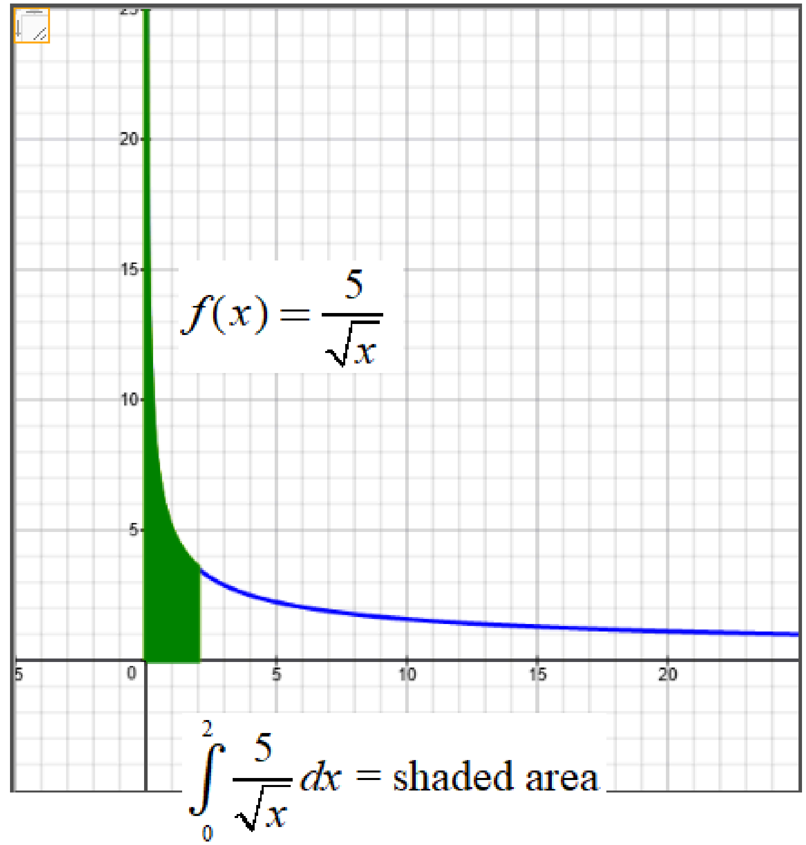


$\displaystyle\int_{-\infty}^0 \frac{1}{e^x}\,dx =$ shaded area

Question 7:

Find $\int_0^2 \frac{5}{\sqrt{x}} dx = \lim_{b \to 0^+} \int_b^2 \frac{5}{\sqrt{x}} dx$

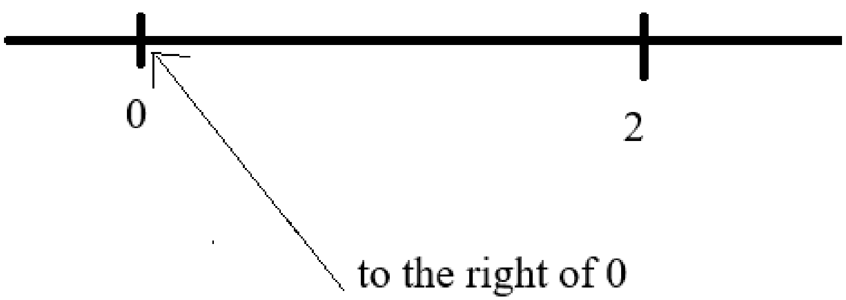
Note: $f(x) = \frac{5}{\sqrt{x}}$ is undefined when $x = 0$.



a) The indefinite integral $\int \frac{5}{\sqrt{x}} dx = 5\int x^{-1/2} dx = ?$

b) $\int_0^2 \frac{1}{\sqrt{x}} dx \approx \int_{0.01}^2 \frac{1}{\sqrt{x}} dx =$ ___?___

c) $\int_0^2 \frac{5}{\sqrt{x}} dx = \lim_{b \to 0^+} \int_b^2 \frac{5}{\sqrt{x}} dx =$ ____________________



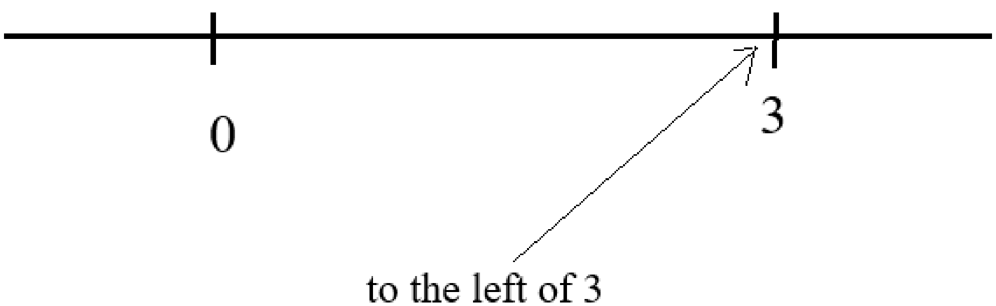
Question 8:

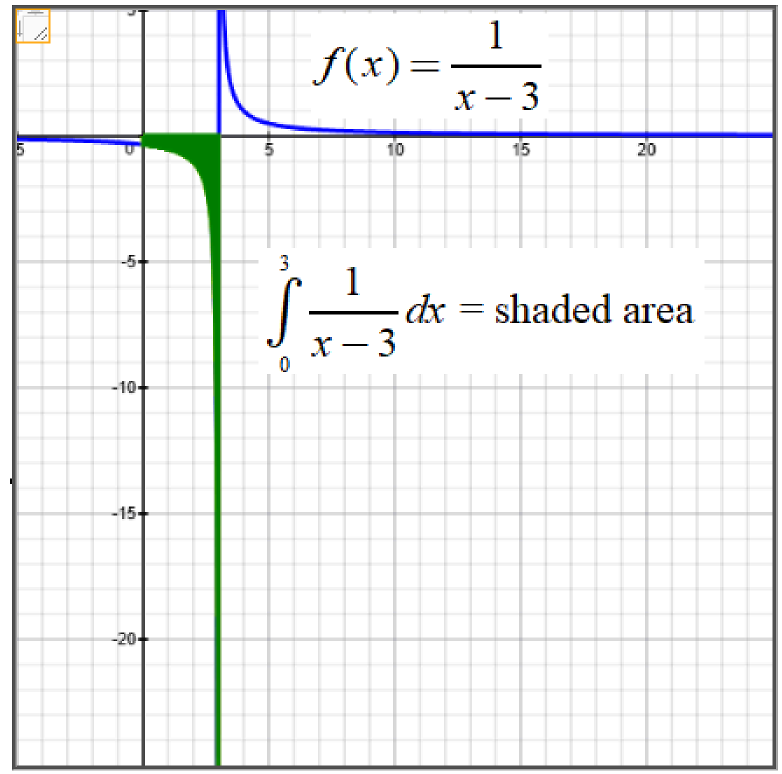
Find $\int_0^3 \frac{1}{x-3}\,dx = \lim_{b \to 3^-} \int_0^b \frac{1}{x-3}\,dx$ Note: Note: $f(x) = \frac{1}{x-3}$ is undefined when $x = 3$.

a) The indefinite integral $\int \frac{1}{x-3}\,dx =$ __?__

b) $\int_0^3 \frac{1}{x-3}\,dx \approx \int_0^{2.999} \frac{1}{x-3}\,dx =$ __?__

c) $\int_0^3 \frac{1}{x-3}\,dx = \lim_{b \to 3^-} \int_0^b \frac{1}{x-3}\,dx =$ __?__





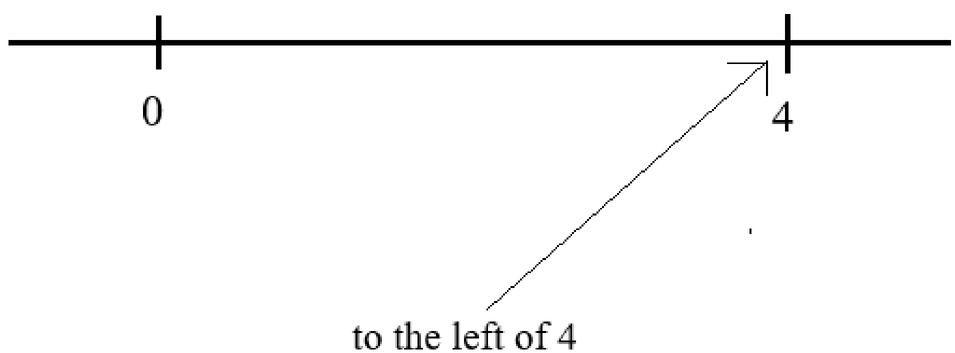
Question 9:

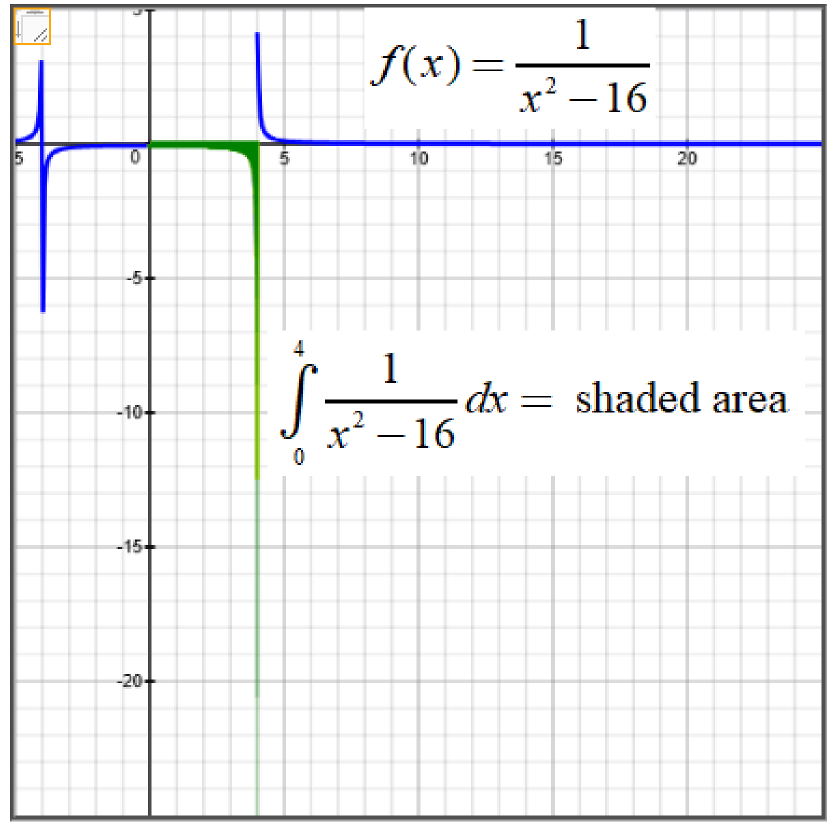
Find $\displaystyle\int_0^4 \frac{1}{x^2-16}\,dx = \lim_{b\to 4^-}\int_0^b \frac{1}{x^2-16}\,dx$ Note: $f(x)=\dfrac{1}{x^2-16}$ is undefined when $x=4$.

a) The indefinite integral $\displaystyle\int \frac{1}{x^2-16}\,dx =$ ___?___

b) $\displaystyle\int_0^4 \frac{1}{x^2-16}\,dx \approx \int_0^{3.999} \frac{1}{x^2-16}\,dx =$ ___?___

c) $\displaystyle\int_0^4 \frac{1}{x^2-16}\,dx = \lim_{b\to 4^-}\int_0^b \frac{1}{x^2-16}\,dx =$ ___?___





Question 10:

Find $\int_2^{\infty} \frac{\ln x}{x}\,dx = \lim_{b\to\infty} \int_2^{b} \frac{\ln x}{x}\,dx$

a) The indefinite integral $\int \frac{\ln x}{x} dx = \int \ln x \cdot \frac{1}{x} \cdot dx = \underline{\quad ? \quad}$ Hint: Let $u = \ln x$

b) $\int_2^{\infty} \frac{\ln x}{x}\,dx = \lim_{b\to\infty} \int_2^{b} \frac{\ln x}{x}\,dx = \underline{\quad ? \quad}$

c) Is the shaded area bounded or unbounded?